Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/20/2010
PI Name:	Dynlacht, Joseph Robert Ph.D.		
Project Title:	Effects of Estrogen on Cataract Induction After Exposure to High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202-5121	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 5/31/2011 per C. Guidry/JSC (10/2010) NOTE: PI to receive NCE to 8/31/2010 per J. Dardano/JSC (7/09)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Mendonca, Marc (Indiana U) Valluri, Shelia (Indiana U) Bigsby, Robert (Indiana U)		
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Task Description:	Several studies have demonstrated that there is an increased cataratogenic potential in animals and humans exposed to high linear energy transfer (LET) radiation compared to low LET radiation. Since the relative biological effectiveness (RBE) of heavy energetic charged (HZE) particles for lens opacification is high, these studies imply a high cataract risk for astronauts involved in long-duration space missions. During a manned mission to Mars, ocular doses in excess of ~0.1 Gy of HZE particles could be accumulated. Though not directly life-threatening, radiation-induced cataractogenesis represents a potentially serious sequelae which could necessitate surgical intervention in the years after, or perhaps even during the mission in some individuals (see below). While the cellular and molecular mechanism(s) of radiation-induced cataractogenesis have not been clearly elucidated, damage to DNA at the time of exposure and subsequent proliferation of the radiosensitive cells in the germinative zone of the lens epithelium likely play a role in the process. Using a rat model, we have recently accumulated data which indicate that estrogen increases or reduces the latent period and may modulate the incidence and severity of cataracts induced by low LET radiation, depending upon when the estrogen is administered. High estrogen levels are induced in non-pregnant women using oral contraceptives, or in post-menopausal women on estrogen replacement therapy. Therefore, sub-groups of female astronauts may be at an increased risk for developing cataracts that are more severe or occur with a more rapid onset. The interaction between estrogen and high LET radiation has never been studied before in the context of cataractogenesis. Therefore, a mechanistic approach characterizing the modulating effect of estrogen on high LET radiation-induced cataractogenesis is proposed. The underlying hypothesis of this proposal is that estrogen alters the latent period for high LET radiation-induced cataractogenesis, and, depending upon when the it is administered, modulates severity and incidence of cataracts. Estrogens regulate several proteins involved in cell cycle control and apoptosis, and it's metabolism results in the production of free radicals which may be genotoxic and mutagenic to mammalian cells. Thus, a novel hypothesis to be tested in the proposed studies is that estrogen alters cell cycle regulation, DNA double strand break (DSB) induction or repair, and proliferation in lens cells exposed to HZE particles (1 GeV/amu iron ion beams). The first Specific Aim shall be to define the dose-time interactions of high LET radiation (vs. low LET radiation) and estradiol to better understand the mechanism of estrogen action. The RBE for cataractogenesis in estrogen-treated animals exposed to1 GeV iron beams shall be determined using 60-Co gamma rays as a standard. The second specific aim shall be to determine the mechanism of estrogen-modulation of radiation-induced cataracts. Animal and cell culture studies will be performed to determine the effects of estrogen on gene expression, cell proliferation, apoptosis, and the induction and repair of radiation-induced DSBs. Slit-lamp biomicroscopy will be used to measure lens opacification in exposed animals. Successful completion of these studies will lead to a greater understanding of the modulation of space radiation effects by hormones and novel intervention strategies for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results obtained from this project may provide greater insight into the potential effects of hormones on normal tissues after radiotherapy treatments with high LET particles.
Task Progress:	This report summarizes progress during a 12-month no-cost extension period for NASA Grant NNJ05HE64G. Work was focused on accruing data from experiments to complete the first Specific Aim of the project [to determine whether estrogen (17-beta-estradiol, E2) alters cataractogenesis induced by high LET radiation]. In all of our experiments, we used our specially designed animal holder and collimator set-up to irradiate the right eyes of up to 3 rats simultaneously, while sparing the contralateral eye and most other normal tissues. This resulted in other structures receiving a dose of <3.5% of the dose to the target eye; no radiation-induced cataracts have ever been observed in the contralateral eye. The results reported in previous Task Reports validated the use of our animal holder and collimator set-up for small volume irradiations. We completed our accrual of data from experiments that was initiated up to nearly two years earlier, in which we compared the incidence and progression of cataracts induced in young vs. old rats with 2 Gy of a 600 MeV/amu 56-Fe ion beam or 10 Gy of gamma rays. Eyes of either intact, ovariectomized and placebo-implanted, or ovariectomized E2-implanted 56-day old or 1 year old female rats were irradiated and slit-lamp biomicroscopy was used to measure lens opacification in exposed animals. Opacification of the anterior and posterior subcapsular (ASC and PSC, respectively) lenses of irradiated and control (contralateral) eyes was measured every 2-4 weeks. In addition to permitting an assessment of age differences for cataractogenesis after high and low LET irradiation, these experiments also allowed us to determine whether estrogen would modulate radiation cataractogenesis in young or old rats. The data we accumulated indicate that the latent period for cataracts is reduced and the incidence is increased in older rats whose eyes were exposed to high LET radiation, compared to younger rats. It is also evident that the type of radiation is an important factor in the age-response of the lens. Our results obtained after irradiation of rat eyes with high LET 56Fe ions contrast with the results from our own low LET experiments. This could be due to a myriad of factors, including the type of damage induced by low vs. high LET, and how the damage is repaired. However, the results of our high LET study are consistent with limited data obtained from studies in which rodents were exposed to radiations of higher LET. Incidence and progression curves were similar for old rats that were either left ovary intact or ovariectomized one week prior to irradiation with iron ions. However, old rats which were ovariectomized but treated with estrogen commencing one week prior to irradiation and continuing throughout the period of observation showed a greater incidence rate. Moreover, incidence in the estrogen-treated group rose to 100% before significant opacification was noted in the intact or ovariectomized groups. These results suggest that exogenously supplied estrogen potentiates radiation cataractogenesis in older rats. This would imply that estrogen status may account for differences in cataractogenesis within a specific age group after irradiation of eyes with 2 Gy. It would also imply that the potentiating effect of estrogen on cataractogenesis induced by high LET radiation is independent of age; although the extent of the potentiating effect of estrogen is lesser in older rats. Finally, our data indicate that estrogen status cannot account for the age-dependent differences in cataractogenesis noted after exposure to 2 Gy of high LET radiation. We continued to accrue data for experiments initiated as early as the Fall 2006 run at NSRL, at which time a radiation dose response experiment was started to determine whether modulation of cataractogenesis by E2 after exposure to high LET radiation is dependent upon the radiation dose. Ovariectomized or intact female rats, or ovariectomized rats that received estrogen-implants were irradiated with 0.05 to 2 Gy of 600 MeV/amu 56Fe ions. Data accrual is almost complete for the 0.5 Gy groups and data for the 1-2 Gy groups was reported in previous Task Reports, but data accrual is continuing for animals irradiated with lower doses. At this time though, it appears that estrogen enhances

	cataractogenesis to a greater extent after exposure to lower doses of iron ions. Upon completion of data accrual for all of these experiments, the issue of age differences in radiation cataractogenesis, and the issue of whether hormones may be determinants, will be better understood. These data may have important implications in regard to age- and gender-related risk issues for extended manned space flight.
Bibliography Type:	Description: (Last Updated: 07/17/2025)
Abstracts for Journals and Proceedings	Henderson MA, Valluri S, Garrett J, Lopez JT, Mendonca MS, Caperell-Grant A, Bigsby RM. "Age and Hormonal Modulation as Determinants of Cataractogenesis Induced by High LET Radiation." Presented at the 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010. Abstract Book. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010. , May-2010
Articles in Peer-reviewed Journals	Dynlacht JR, Valluri S, Garrett J, Mendonca MS, Lopez JT, Caperell-Grant A, Bigsby RM. "Age and Hormonal Status as Determinants of Cataractogenesis Induced by Ionizing Radiation. I. Densely Ionizing (High LET) Radiation." Radiation Research. (In Press). 2010. , Oct-2010
Articles in Peer-reviewed Journals	Henderson MA, Valluri S, Garrett J, Lopez JT, Caperell-Grant A, Mendonca MS, Rusek A, Bigsby RM, Dynlacht JR. "Effects of estrogen and gender on cataractogenesis induced by high-LET radiation." Radiat Res. 2010 Feb;173(2):191-6. PMID: 20095851 , Feb-2010